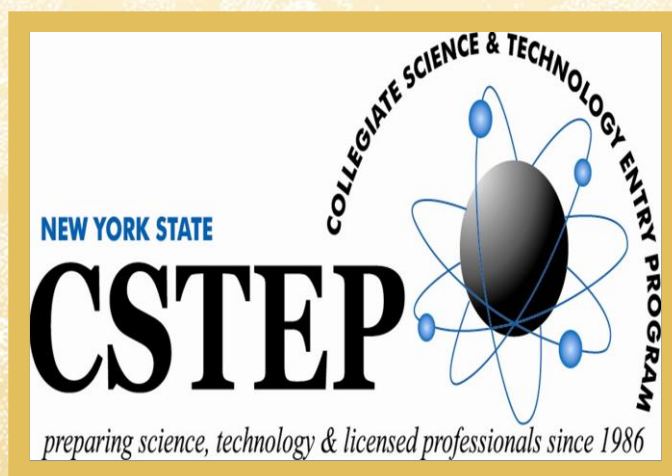


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SMART METER HOME APPLICATION

A PROPOSAL TO: Dr. SOVAK



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INTRODUCTION

Home Automation is one of the most famous application profiles that is being researched and implemented by many companies around the world. ZigBee is the global and rapid grow standard of Wireless Sensor Network. This allows the smart meter to communicate with several electrical appliances within the same network. The ZigBee is built on IEEE 802.15.4 radio and MAC and it is the only standard (until now) that support wireless mesh network. Within ZigBee standard, numerous “application profiles” define the standard ZigBee devices that could be used to communicate with each other.

The goal of our project is to design a Smart Meter Application using ZigBee. The Home Automation Smart Meter Application will be able to monitor the amount of energy consumed by users. This application places ZigBee module into our daily used devices such as lamp, radiator, ventilation, and others. The Smart Meter Application monitors every device by judging from the read out from the sensor. Simple devices like those mentioned above could easily be combined together to form a large and complex network within a home. This network will not only make life more comfortable but also save energy dramatically. Likewise, users will be able to see how much energy is being used during peak hours when the cost of energy is high. Therefore, the user will be allowed to manage their consumption conveniently and in detail. The amount of energy used by specific appliances will be monitored and when necessary the appliance can be shut off. The Smart Meter Application will also be able to automatically shut off an appliance when too much energy is being used. In brief, the smart meter application makes it possible for users to conserve electricity and save money.



ABSTRACT

04/24/2012

Energy meters measure the amount of electric energy consumed and are typically calibrated in billing units and measured in kilowatt hours. The term digital refers to time sampling systems, where analog to digital converter (ADC) translates real world waveforms to binary words that digital circuits understand and manipulate. In digital energy meters, voltage and current waveforms in the actual circuit are sensed using transducers and transformed to smaller values that are sampled and converted to binary words. Once converted to a digital signal voltage and current are multiplied and integrated over time to extract energy consumption.

One of the main features of this energy meter is two current transducers that measure active power in live and neutral wires. This enables the meter to detect, signal, and continue to measure reliably even when tampered with. The project designs and programs circuit peripherals according to microcontroller specifications. The program on the microcontroller multiplies incoming voltage and currents in time domain and calculates the total signal power. The digital energy meter will be calibrated using reference voltage and current waveforms.

“Save Energy while Controlling and Protecting your Electronic Appliances”

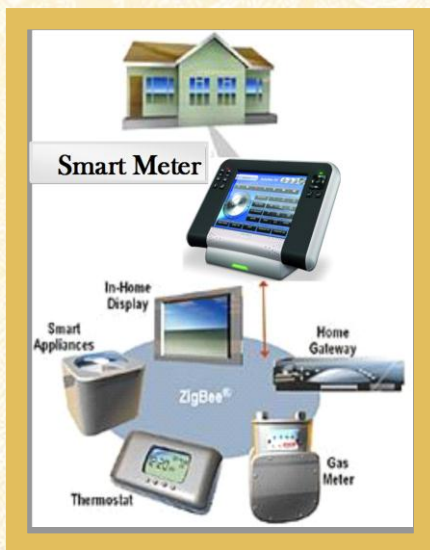


Figure 1

Affordable Technology

The wireless smart meter application is a cheap and easy way to provide network communication and power control between electronic devices at places with no wired infrastructure. The smart meter wireless communication provides all suitable data rates as compared to other wireless technology so this withstands suitably for all applications. This new technology also provides reliable protection for the operation of electric power systems and benefits to industrial automation applications because of low cost deployment and redeployment.

ZigBee Technology

A ZigBee network, in a home or industry, can easily link electronic devices to save energy and money while protecting all electronic devices in the network. Home automation application will be able to monitor the amount of energy that users consume through ZigBee modules that will be receiving power signals from daily used devices such as lamp, radiator, ventilation, and others.

This smart meter connected to a ZigBee module to monitor and combine all electronic devices to form a large and complex network within a home or industry. Through this network, the smart meter will shout any device off when needed as well as when too much energy is used. The user will be able to manage their consumption conveniently and in detail. In fact, ZigBee network technology will not only make life more comfortable but also save energy and money while protecting your environment.

As part of a “Smart-Grid” project, the Digital Energy Meter within the Smart meter interfaces to the grid and helps customers reduce energy consumption. With this technology, customers can access information on consumption patterns and reduce usage during peak hours.

HOME ENERGY CONSUME PROBLEMS

Where is energy often used and wasted!

The energy bills for a typical US family home totals approximately \$2,200 per year.

Where does all this money go?

About half of this spend is on just heating and cooling the air within the home, which is why weatherization and energy efficient heating & cooling systems are so important to lower your bills. The other half is spent on heating water and all of the large and small appliances, lighting, and electronics throughout the home.

The chart below shows the breakdown of energy use by category and gives a sense of where savings can be found.

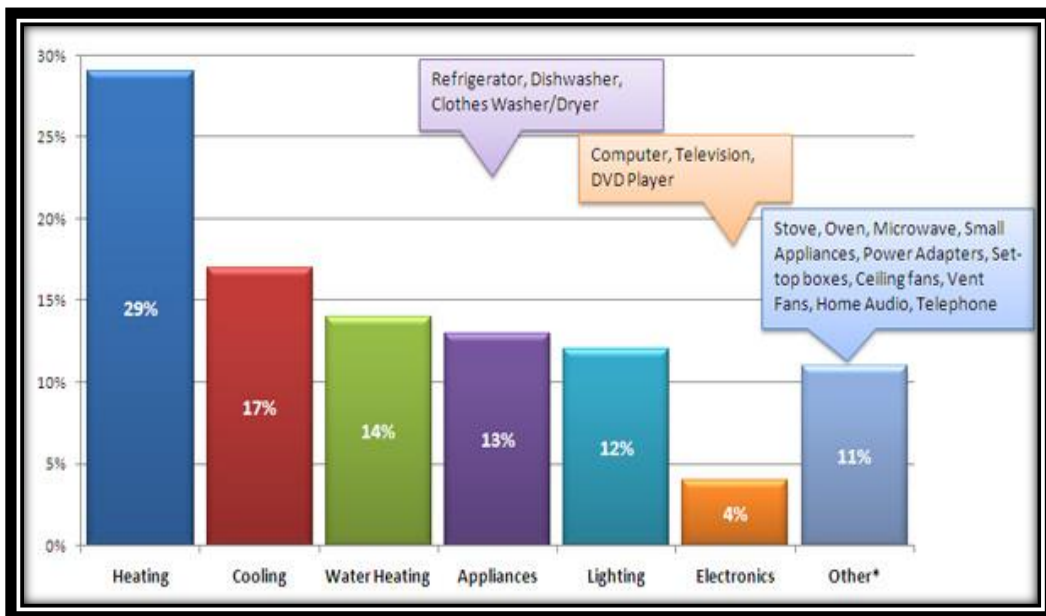


Figure 2

HOME ENERGY CONSUME SOLUTIONS

Inspection: Taking a few minutes once a month to examine your appliances to reveal any possible problems. Check for electrical cords cracks and also any other obvious damage.

Batteries: It is helpful to keep a list to let you know when you last replaced your appliances batteries.

Checking Surge Protectors: A wonderful tool to keep your appliances from getting damaged through a power surge; this is very important when you are talking about things like televisions and computers. Making sure that they are working properly can spare you the loss of the appliance from a power surge.

Upgrades: Sometimes we rely upon an appliance that is outdated and consuming a great deal of energy. It can save us money on electrical bills by replacing them with a more efficient model.

Checking Outlets: It is important to periodically make sure that you outlets are working correctly, because if one stops working or is faulty it can affect your appliance.

Turning Off Appliances: Making sure we turn off something, like a computer that is not in use, can save energy and cut down on possible problems from it being on for too long of a period, when not in use.

What if we had one only electronic device that does the above solution methods for us?

Yes, the Smart Meter Application based on a ZigBee network can do each one of these functions while saving the users energy and money. The smart meter network, when placed into our daily used electronic appliances, can easily combine them together to form a simple network within a home and use them efficiently.

SMART METER APPLICATION & ZIGBEE TECHNOLOGY

The Smart Meter uses ZigBee protocols built in low power consume technology. Every application consumes the least possible amount of energy by maintaining all synchronized electronic devices within the same network in sleep mode.

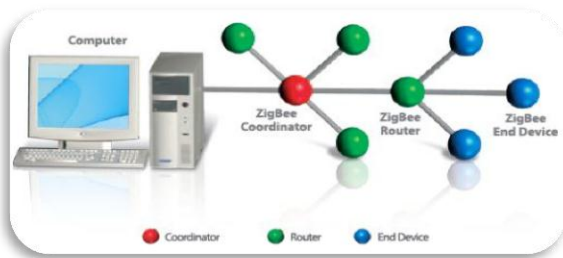


Figure 3

ZigBee network layer supports different application methods such as star, tree, and mesh topologies. In the star topology, the network is controlled by one

single device called the ZigBee coordinator. The ZigBee coordinator initiates the communication and maintains end-devices on the network.

In tree networks, routers move data and control messages through the network using a hierarchical routing strategy. And in the mesh and tree topology the ZigBee coordinator starts the network and chooses certain key network parameters, network may be extended through the use of ZigBee routers. Mesh networks perform a complete peak-to-peak communication and it is described only by intra-PAN networks through which communications start and end within the same network.

SMART METER APPLICATION & ZIGBEE TECHNOLOGY

This Smart Meter Application design is based on a low-cost and low-data rate wireless technology that consists of three main devices to incorporate the ZigBee radio application.

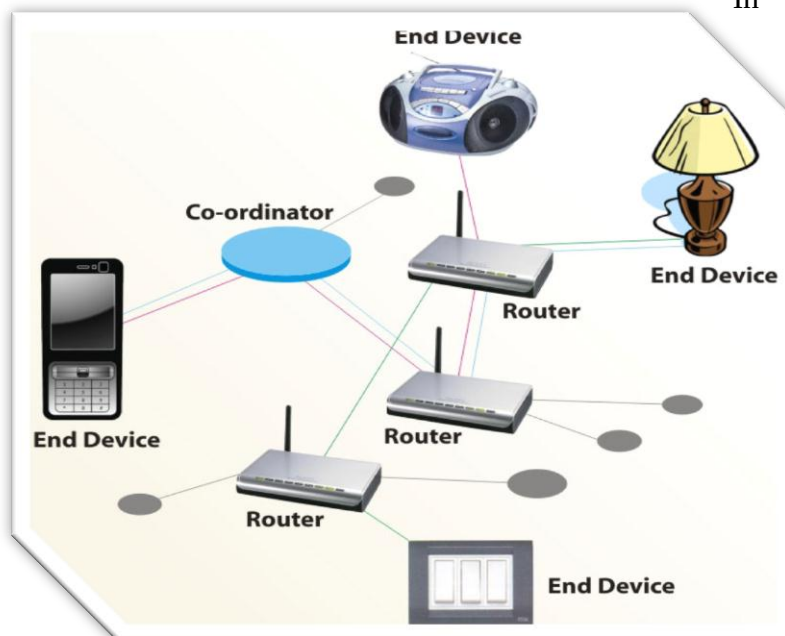


Figure 4

In a ZigBee specification are found three kinds of devices that incorporate ZigBee radios, and these are the followings. A coordinator: this organizes the network and maintains routing tables. Routers: this can talk to the coordinator, to other routers and to

reduced-function end devices. Reduced-function end-devices: this can talk to routers and the coordinator, but not each other.

Routers and the coordinator are mains-powered and do not go to sleep while end devices wake up only when they need to communicate and immediately go back to sleep to minimize power consumption and promote long battery life in battery-powered devices.

HARDWARE DEVELOPMENT AND PROCEDURES

Setting up the SK40C and PIC as a coordinator and router simulator

Setup SK40C with PIC. First prepare a PCB board and all the components required according to the schematic on the right.

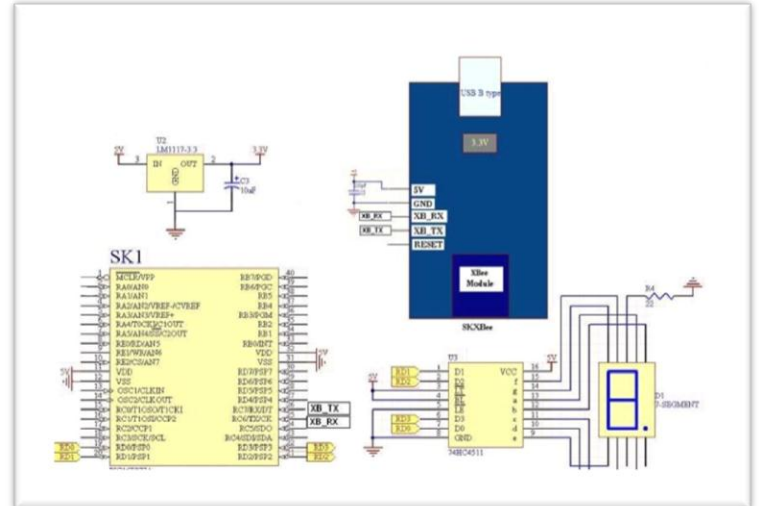


Figure 5

Load programs to the microcontroller using a PCI interface.

Sample source codes are downloaded from the web.

```
void init(void) // subroutine to initialize
{
    SPBRG=129; // set baud rate as 9600 baud
    BRGH=1;
    TXEN=1;
    CREN=1;
    SPEN=1;
    TRISD = 0b00000000;
    seg = 0b00000000;
}
```

Figure 6

Figure 6

Turn on, press and release reset
button of SK40C.

After setting everything up, the SK40C will be able to receive data through the network.

Using SKXBee as a Coordinator

Connect SKXBee to PC using USB cable as the left picture. Set the address for each module, test the communication and go to ‘Terminal’ setting tab.

Press reset button of SK40C, followed by pressing “ok” and “Press any number” will appear.



Figure 7

Connect end devices. Red color is for data received while blue color is for data sent.

Energy consumed will appear in the LCD screen.

COMPARISON & BENEFITS

COMPARISON

The Smart Meter Home Application like any other new smart meter in the market measures the total amount of energy consumed in a home. However, unlike other smart meters this smart meter application can drastically save the user more energy and money. According to our research, other new meter installations can cost about \$10,104 in total while the smart meter home application can be installed for much less than a \$100 dollars.

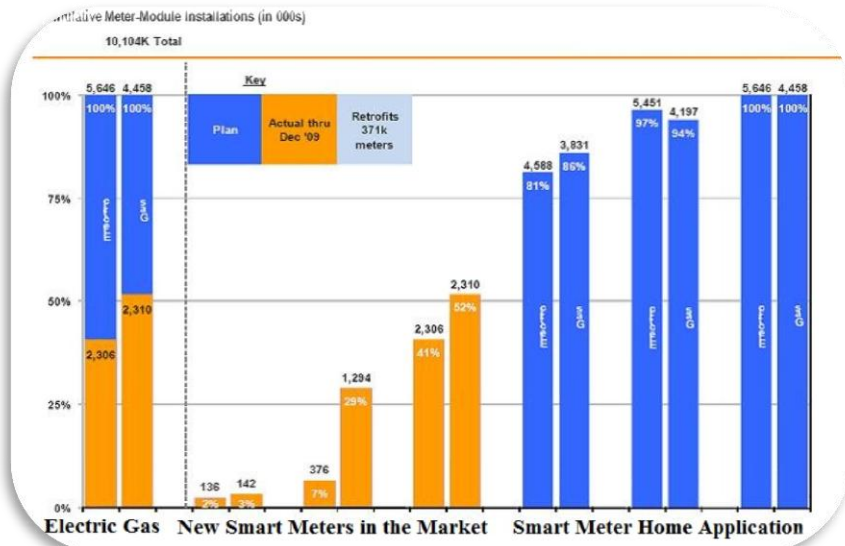


Figure 8

BENEFITS

The Smart Meter Home Application also can be programmed to save energy during peak hours. The chart to the left shows how much energy and money it can drastically save by automatically turning off electronic appliances not needed during peak hours. Moreover, this programming feature can be even more beneficial by allowing the user to be able to turn on/off electronic devices wirelessly while being away from home. In the same way, air

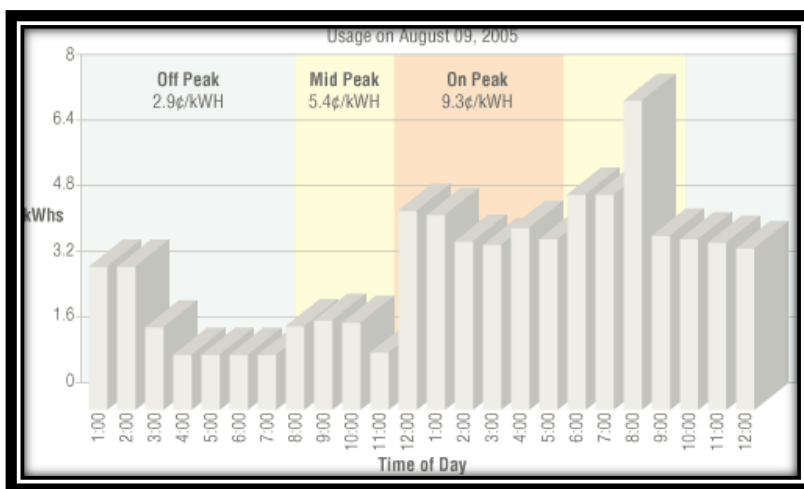


Figure 9

conditioner and heaters will be monitored and controlled by software that will simulate a thermal status to maintain the home environment in a temperature that the user desires while saving energy.

CONCLUSION

Wireless communication is a cheap and easy way to provide network communication at places where there is no wired infrastructure. ZigBee provides all suitable data rates as compared to other wireless technology and withstands suitably for all applications. ZigBee technology provides reliable protection for the operation of electric power systems and benefits for home and even industrial automation applications because of low cost deployment and redeployment. The Smart Meter ZigBee network application can be placed into our daily used devices such as lamp, radiator, ventilation, and others. This Application can monitor every device by judging from the read out from the sensor. Devices like those mentioned could easily be combined together to form a large and complex network within a home and even an industry.

The Smart Meter Home Application does not only make life more comfortable but also save energy. The amount of energy used by specific appliances is monitored and when necessary the appliance is shut off. This application is also able to automatically shut off an appliance when too much energy is being used. In brief, the ZigBee power monitoring system application makes it possible for users to conserve electricity and save money.

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